Government College of Engineering Aurangabad,

Chhatrapati Sambhajinaar

Department of Computer Science & Engineering.

Class Test-I, AY-2024-25, SEM-IV (NEP)

Class: SYCSE **Subject:** Data Structure **Course Code:** CSPCC2002

Mark: 15 **Time:** 1Hr

Attempt any three questions.
Each question carries FIVE marks.

		CO	BT
Q. 1	Discuss data structure and it's all types' details.	CO1	BT
Q. 2	Elaborate the Queue and it's sequential representation with suitable example.	CO1	BT
Q. 3	Demonstrate the drawback of sequential storage in data structure and discuss the Linked list with suitable example.	CO2	BT
Q. 4	Illustrate the doubly Linked List and note down the application of it over the conventional linked list.	CO2	BT
Q. 5	Write a program using ADT to perform operations on rational numbers that we define the creation of a rational number from two integers, addition, multiplication, and testing for equality	CO1	BT

	b) It is easy to insert and delete elements in Linked List c) Random access is not allowed in a typical implementation of Linked Lists d) The size of array has to be pre-decided, linked lists can change their size any time. e) All of the above			
Q.3	**Attempt any two**	*CO*	*B.T. Level*	*Marks*
A	Elaborate the Operations on binary tree and application of binary tree in details.	*3*	*L3*	*06*
B	Discuss the Binary tree and their different traversal methods with suitable example.	*3*	*L2*	*06*
C	Define B-Tree of order n, and discuss the B-Tree with considering different insertion of element in to it.	*3*	*L2*	*06*
Q.4	**Attempt All**	*CO*	*B.T. Level*	*Marks*
A	Discuss the Graphs and Defining a traversal that relates to the structure of a graph is more complex than for a list or a tree for three reasons	*4*	*L3*	*06*
B	Discuss the Depth First Traversal technique and application of DFS.	*4*	*L2*	*06*
Q.5	**Attempt All**	*CO*	*B.T. Level*	*Marks*
A	Design the Merge Sort algorithm and elaborate a suitable example on it.	*5*	*L3*	*06*
B	Discuss different searching techniques and elaborate of any one technique in details.	*5*	*L2*	*06*

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level

Government College of Engineering Aurangabad, Chhatrapati Sambhajinagar

(An Autonomous Institute of Government of Maharashtra)

End Semester Examination December 2024

Second Year B. Tech. Computer Sci. & Engineering – (NEP)

CSPCC2002: Data Structures

Time: 2.30Hours 16 DEC 2024 *Max. Marks: 60*

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. *Attempt all questions.*
2. *Figures to the right indicate full marks.*
3. *Assume suitable data if necessary and state it clearly.*
4. *Use of Non-Programmable Calculator and Data Sheet is allowed.*

Q.1	**Attempt any two**	*CO*	*B.T. Level*	*Marks*
A	Elaborate the concept of stack as an ADT in details.	*1*	*L2*	*06*
B	Discuss the concept of Infix, Postfix and Prefix with suitable example.	*1*	*L2*	*06*
C	Illustrate the queue as an ADT with implementation of it.	*1*	*L3*	*06*
Q.2	**Attempt any two**	*CO*	*B.T. Level*	*Marks*
A	Demonstrate the algorithm Insertion and deletion of nodes in linked list at the end of list.	*2*	*L3*	*06*
B	Discuss the Circular Linked List with appropriate example.	*2*	*L2*	*06*

C	Choose the correct option of following and give explanation on it.
	
	i) What does the following function do for a given Linked List with first node as head?
	`void fun1(struct node* head)`
	`{`
	`if(head == NULL)`
	`return;`
	
	`fun1(head->next);`
	`printf("%d ", head->data);`
	`}`
	(a) Prints all nodes of linked list
	(b) Prints all nodes of linked list in reverse order
	(c) Prints alternate nodes of Linked List
	(d) Prints alternate nodes in reverse order
	
	ii) What should be added in place of "/*ADD A STATEMENT HERE*/", so that the function correctly reverses a linked list.
	(a) *head_ref = prev;
	(b) *head_ref = current;
	(c) *head_ref = next;
	(d) *head_ref = NULL;
	
	iii) Which of the following points is/are true about Linked List data structure when it is compared with array?
	a) Arrays have better cache locality that can make them better in terms of performance.
		2	*L3*	*06*

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Class Test-II, AY-2024-25, SEM-III

Class: SYCSE
Mark: 15

Subject: Data Structure

Course Code: CSPCC2002
Time: 1Hr

Attempt any three questions.
Each question carries FIVE marks.

		CO	BT
Q. 1	Distinguish binary tree and non-binary tree and describe the binary tree with suitable example.	CO3	BT3
Q. 2	Discuss the different types of binary tree traversal method, describe it with suitable example.	CO3	BT2
Q. 3	Illustrate the B-tree of order of n and describe the functioning of it with an example.	CO3	BT3
Q. 4	Describe the graph and defining a traversal that relates to the structure of a graph is more complex than for a list or a tree for whichthree reasons.	CO4	BT2
Q. 5	Discuss the Depth-First Traversal with suitable example.	CO4	BT2